

Vectors

Ek arrow jo direction aur magnitude dono batata hai

01 Opening Hook

STORY TIME

Socho tum ek treasure hunt pe ho. Map par likha hai: '5 kadam east, 3 kadam north chalo'. Yeh instruction sirf ek number nahi hai – isme direction bhi hai aur distance bhi. ****Yahi cheez hai ek Vector.****

02 Why Does This Concept Exist?

THE PROBLEM

Agar hum sirf ek number (jaise '5') bolte, toh pata nahi chalta kis taraf jaana hai. Real duniya mein bahut si cheezein – force, velocity, ya AI mein 'meaning' – sirf magnitude se describe nahi hoti, unki direction bhi zaroori hoti hai. Vector isliye invent hua taaki hum 'kitna' aur 'kis taraf' dono ek saath describe kar sakein.

03 Explain Like I'm 10

SIMPLE WORDS

Socho ek arrow hai jo ek point se doosre point ki taraf ja raha hai. Arrow ki lambai batati hai 'kitna' (magnitude), aur arrow kis taraf ishara kar raha hai woh batata hai 'kis taraf' (direction).

School mein bhi tumne coordinates dekhe honge (x,y) – jaise $(3,4)$ matlab 3 right, 4 up. Yeh bhi ek vector hai – starting point $(0,0)$ se $(3,4)$ tak ka arrow.

04 Build Intuition First (No Maths Yet)

FEEL IT FIRST

Vector ko ek 'list of numbers' ki tarah bhi soch sakte ho, jahan har number ek direction mein movement batata hai. AI mein hum har cheez ko - word, image, user preference - ek vector (numbers ki list) mein badal dete hain, taaki hum unke beech 'distance' ya 'similarity' calculate kar sakein.

05 Ab Thoda Maths

FORMULA TIME

Vector Notation:

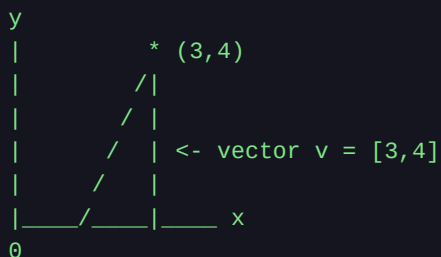
Vector ko likhte hain jaise $v = [3, 4]$ (2D) ya $v = [x_1, x_2, \dots, x_n]$ (n-dimensional). Iski magnitude (length) hoti hai $\|v\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$.

Vector Addition:

Do vectors ko add karna matlab unke corresponding components ko add karna: $[1, 2] + [3, 1] = [4, 3]$. Geometrically, ek arrow ke end se doosra arrow shuru hota hai.

06 Visualize Everything

PICTURE IT



Word Embedding bhi ek vector hai:
"king" -> [0.21, -0.5, 0.83, ...] (300+ dimensions!)

07 AI Connection

WHERE AI USES THIS

Word Embeddings: Har word ko GPT/Claude jaise models ek high-dimensional vector mein represent karte hain (jaise 768 ya 4096 numbers). Similar meaning wale words ke vectors close hote hain.

Image Vectors: Ek image ko bhi CNN ek vector (feature vector) mein compress kar deta hai jo uski visual properties encode karta hai.

08 Real Industry Example

REAL COMPANIES

OpenAI: Text embeddings API text ko high-dimensional vectors mein convert karti hai jo semantic search aur RAG systems mein use hote hain.

Pinecone/Weaviate: Vector databases specially banaye gaye hain taaki millions of vectors mein fast similarity search ho sake.

09 Explain Like I'm Going To Build It

HOW YOU'D CODE IT

PyTorch mein ek vector sirf ek 1D tensor hai: `torch.tensor([1.0, 2.0, 3.0])`. Embedding layers (`nn.Embedding`) har word/token ko ek vector mein map karte hain jo training ke dauraan seekha jaata hai.

10 Common Misconceptions

BEGINNER MISTAKES

Galat soch: Vector aur point same hote hain.

Sahi baat: Point ek location hai, vector ek 'movement' ya 'direction+magnitude' hai - lekin origin se draw karne par vector ek point jaisa dikh sakta hai.

Galat soch: Zyada dimensions matlab zyada confusing.

Sahi baat: High-dimensional vectors (jaise 1536-dim embeddings) bas 'zyada properties' encode karte hain - concept wahi rehta hai jo 2D vector ka hai.

11 Interview Questions

INTERVIEW CORNER

[Beginner] Q: Vector kya hota hai?

A: Ek quantity jiski magnitude (size) aur direction dono hoti hai, numbers ki ordered list se represent hoti hai.

[Intermediate] Q: Word embeddings vectors kyun hote hain, single numbers kyun nahi?

A: Kyunki word ka meaning multiple properties (context, sentiment, topic) mein hota hai - ek single number itni richness capture nahi kar sakta, isliye high-dimensional vector chahiye.

[Advanced] Q: Do embedding vectors ki similarity kaise measure karte hain?

A: Cosine similarity ya dot product se - jo angle/alignment measure karta hai, magnitude ko ignore karke.

12 Exercises

PRACTICE TIME

[Easy] Vector $[3,4]$ ki magnitude nikalo.

Solution: $\sqrt{3^2+4^2} = \sqrt{25} = 5$.

[Medium] $[2,3]$ aur $[1,-1]$ ko add karo.

Solution: $[3, 2]$.

[Hard] Agar ek 3D vector $[1,2,2]$ hai, uski unit vector (magnitude 1) nikalo.

Solution: Magnitude = $\sqrt{1+4+4}=3$. Unit vector = $[1/3, 2/3, 2/3]$.

13 Golden Summary

Concept	Easy Meaning
Vector	Direction + magnitude wali quantity, numbers ki list se represent
Magnitude	Vector ki 'length', $\sqrt{\text{sum of squares}}$
Embedding	AI mein kisi cheez (word/image) ko represent karne wala vector

Vector Addition	Components ko add karke naya vector banana
Unit Vector	Magnitude 1 wala vector, sirf direction batata hai

14 One-Line Memory Trick

“Vector ek arrow hai – kitna aur kis taraf, dono batata hai.”